SMART ENVIRONMENTAL MONITORING: IoT AND DUST SENSORS IN GEOSPATIAL IMPACT STUDIES

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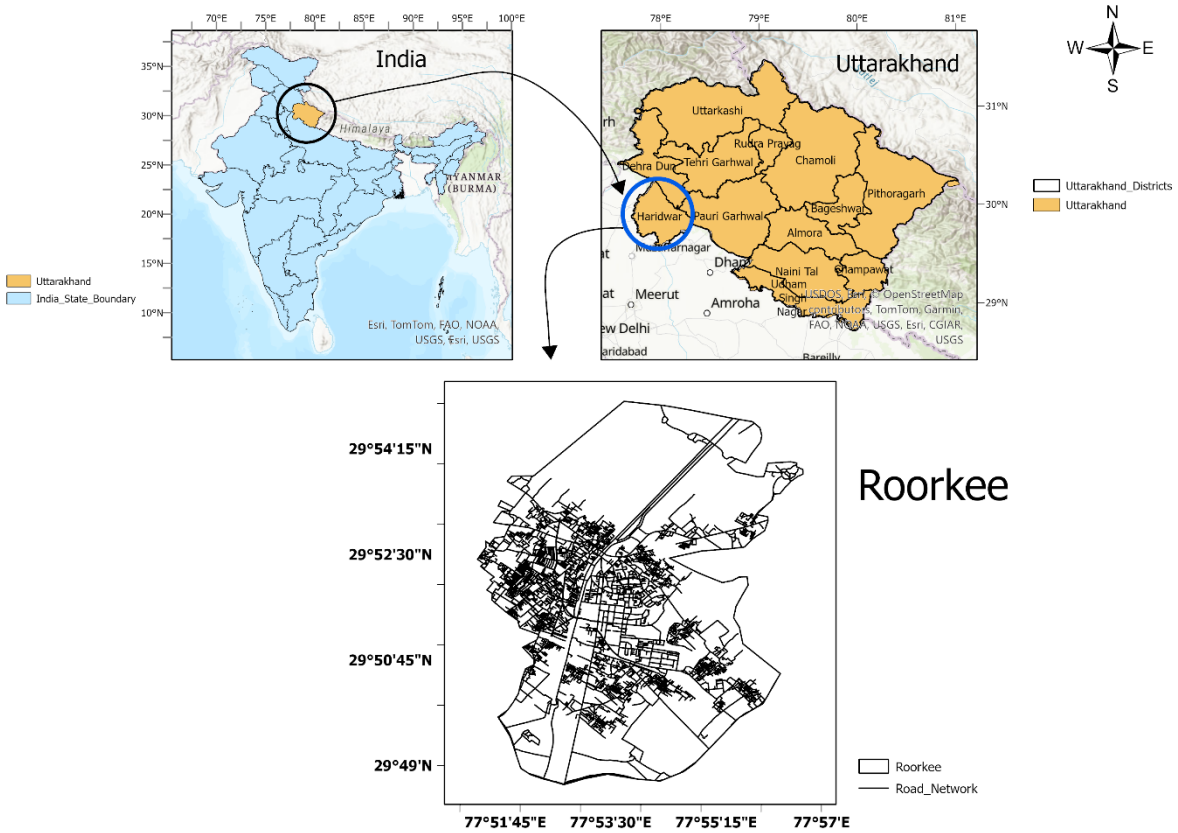
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Problem Statement

- India among top global hotspots for dust-related air pollution. Over 140 million people exposed to hazardous particulate matter (PM).
- **Health Impacts:** Respiratory Diseases, cardiovascular problems.
- **Key Pollution Sources:** Construction activities, industrial emissions, vehicular traffic, natural dust storms.
- **Limitations of traditional problems:** Sparse-static monitoring stations, low spatial and temporal resolution, high operational cost.
- **Need for Innovation:** Real-time, high-resolution monitoring, scalable and cost-effective technology solutions.

Study Area



- Roorkee combines dense urban settlements with high industrial activities.
- Located near Himalayan foothills, high activity of tourists throughout the year.
- Roorkee serves as a model micro-climate to study the interaction of anthropogenic emissions (traffic, construction) with natural dust dynamics.

Conceptual Framework

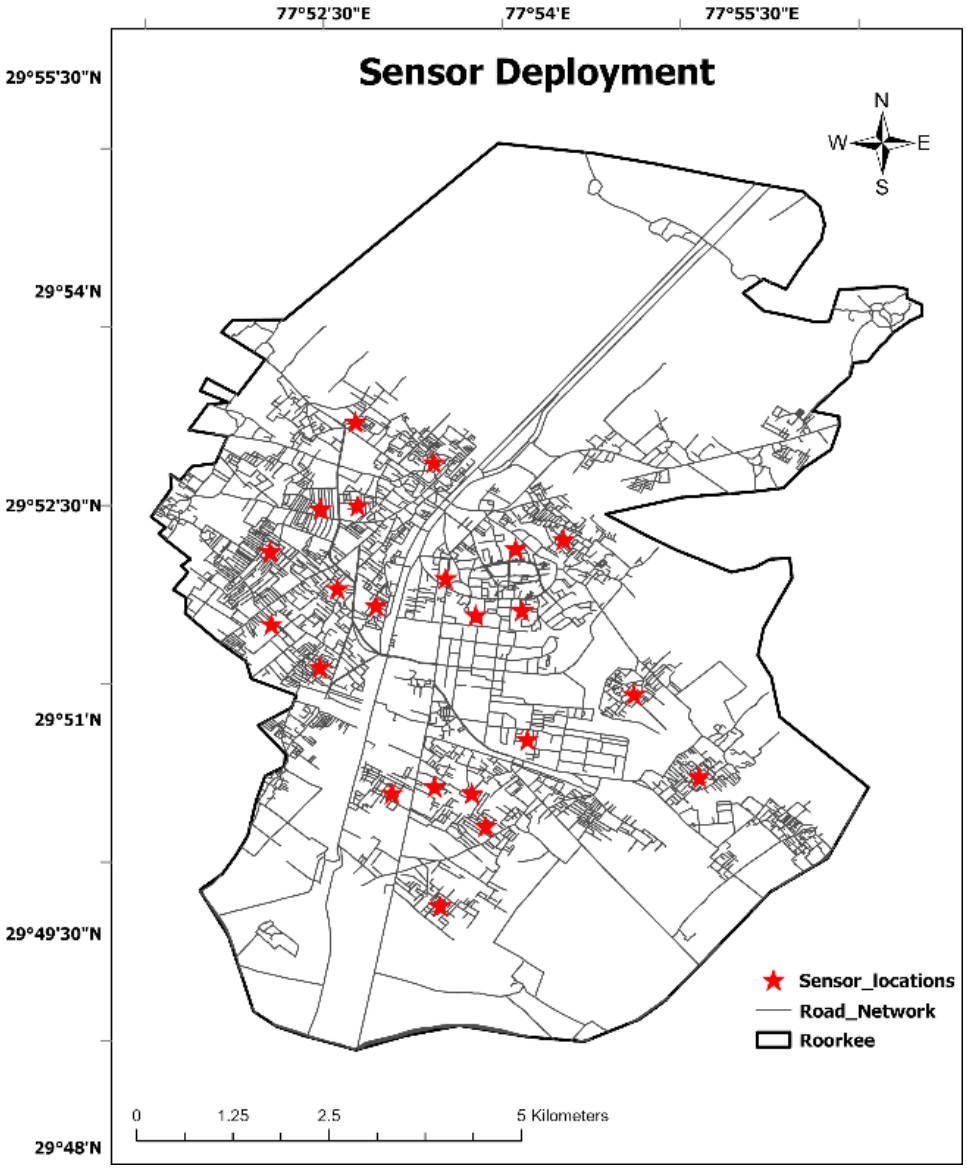
- **IoT-Enabled Dust Sensors:** Low-cost, portable sensors deployed across urban, industrial, and rural zones for real-time particulate monitoring (MQ series).
- **Satellite Intelligence via Sentinel-5P:** TROPOMI instrument provides high-resolution data on dust and co-pollutants (NO_2, CO, CH_4, SO_2) across large areas.
- **Geospatial Analytics through GIS:** Integration of ground and satellite data enables pollution heatmaps, hotspot zoning, and susceptibility analysis.
- **Complementary Strengths:**
  - IoT = granular, real-time, ground truth
  - Satellite = macro-scale spatial trends

….Conceptual Framework

- **Machine Learning for Pattern Detection:** Real-time sensor data processed using ML algorithms to identify trends, anomalies, and peak pollution periods.
- **Supports Policy and Planning:** Framework enables targeted interventions, urban planning improvements, and adaptive air quality management.
- **Outcome:**

A powerful, scalable model for evidence-based environmental monitoring and decision-making.

Sensor Deployment



- Locations throughout Roorkee where sensors were deployed.
- City's extent was covered so that the spatial dynamics continuity was maintained.

Geospatial Analytical Approach

- **Dual-Source Data Integration:** Combined Sentinel-5P satellite data with IoT sensor readings for multi-scale spatial coverage.
- **Hotspot Identification:**
  - Used spatial interpolation (e.g., kriging) and spatial autocorrelation (Moran's I = 0.68) to detect consistent high-dust zones.
- **Temporal-Spatial** Trends: Analysis revealed peak dust periods (e.g., 8–11 AM and 5–8 PM) and weekday–weekend pollution variation.
- **Land Use Correlation:** Mapped pollution hotspots against urbanization, traffic density, and industrial zones to identify key pollution drivers.

….Geospatial Analytical Approach

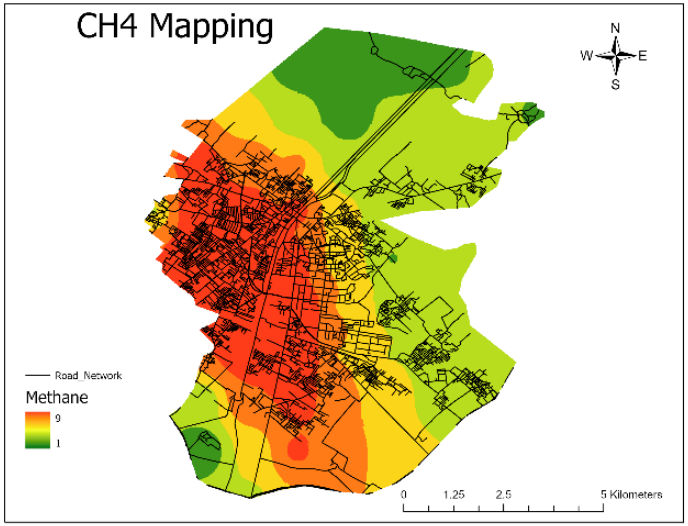
- **Susceptibility Mapping:** Incorporated variables like vegetation cover, wind direction, proximity to industrial areas to assess vulnerability.
- **GIS Visualization:** Generated dynamic pollution heatmaps, trend surfaces, and risk zones for targeted intervention planning.

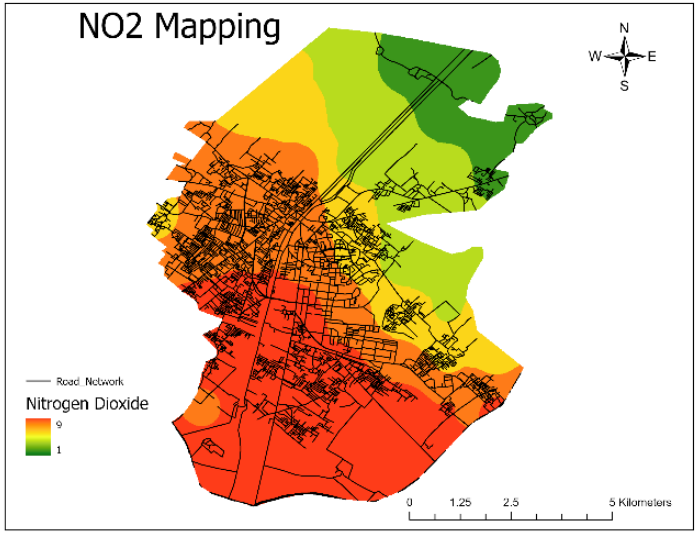
Sentinel-5P Satellite Data: Macro-Level Atmospheric Monitoring

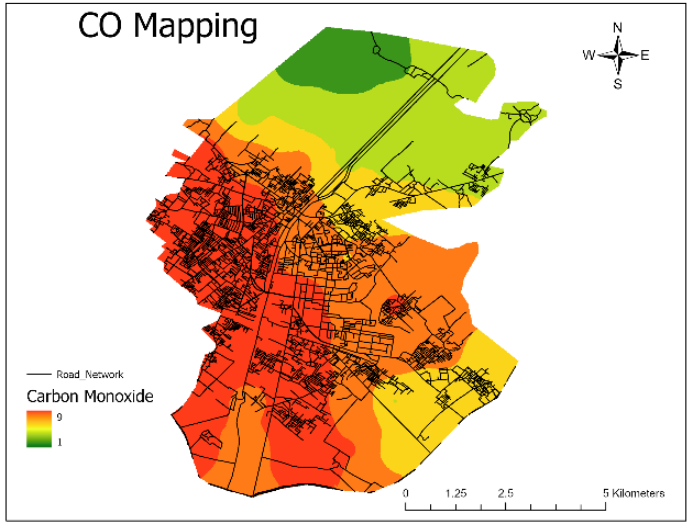
- **TROPOMI Instrument:** Captures high-resolution data on atmospheric pollutants: NO_2, CO, SO_2, CH_4, and aerosols (dust) with daily global coverage.
- **Large-Scale Pollution Mapping:** Enables regional and national-level visualization of dust and gaseous pollutants across time.
- **Geospatial Alignment:** Satellite data georeferenced and integrated with IoT sensor locations using GIS platforms for enhanced spatial analysis.
- **Identifies Preliminary Hotspots:** Detects persistent high-pollution zones even in remote or sensor-sparse areas.
- **Validates and Complements Ground Data:** Cross-verification with IoT sensor readings ensures higher accuracy and consistency in environmental monitoring.

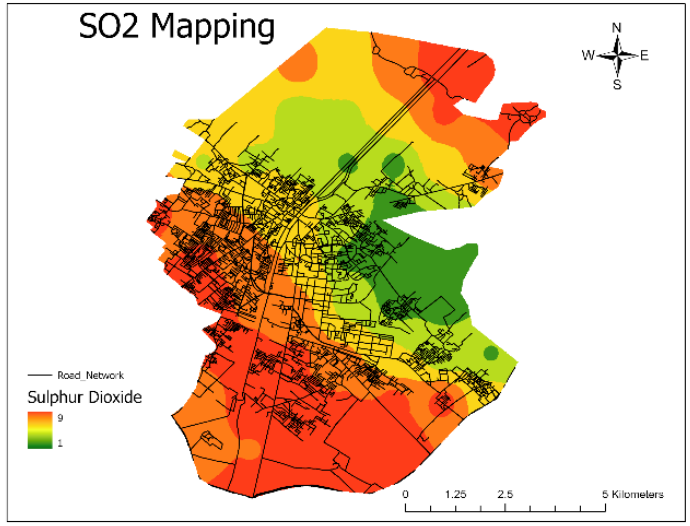
Results and Findings

Mapping from **Sentinel-5P** Data Analytics









Mean Sensor Readings over a month's Monitoring

Sensor 1	0.038
Sensor 2	0.039
Sensor 3	0.034
Sensor 4	0.104
Sensor 5	0.02
Sensor 6	0.08
Sensor 7	0.09
Sensor 8	0.11
Sensor 9	0.063
Sensor 10	0.04
Sensor 11	0.099
Sensor 12	0.105
Sensor 13	0.045
Sensor 14	0.057
Sensor 15	0.09
Sensor 16	0.12
Sensor 17	0.06
Sensor 18	0.08
Sensor 19	0.064
Sensor 20	0.124
Sensor 21	0.059
Sensor 22	0.096

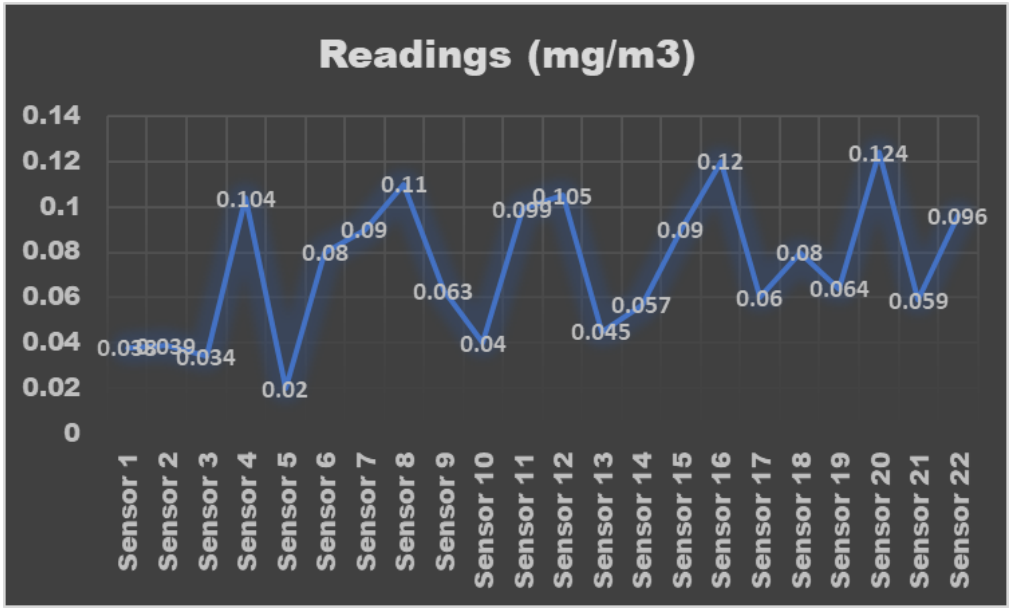
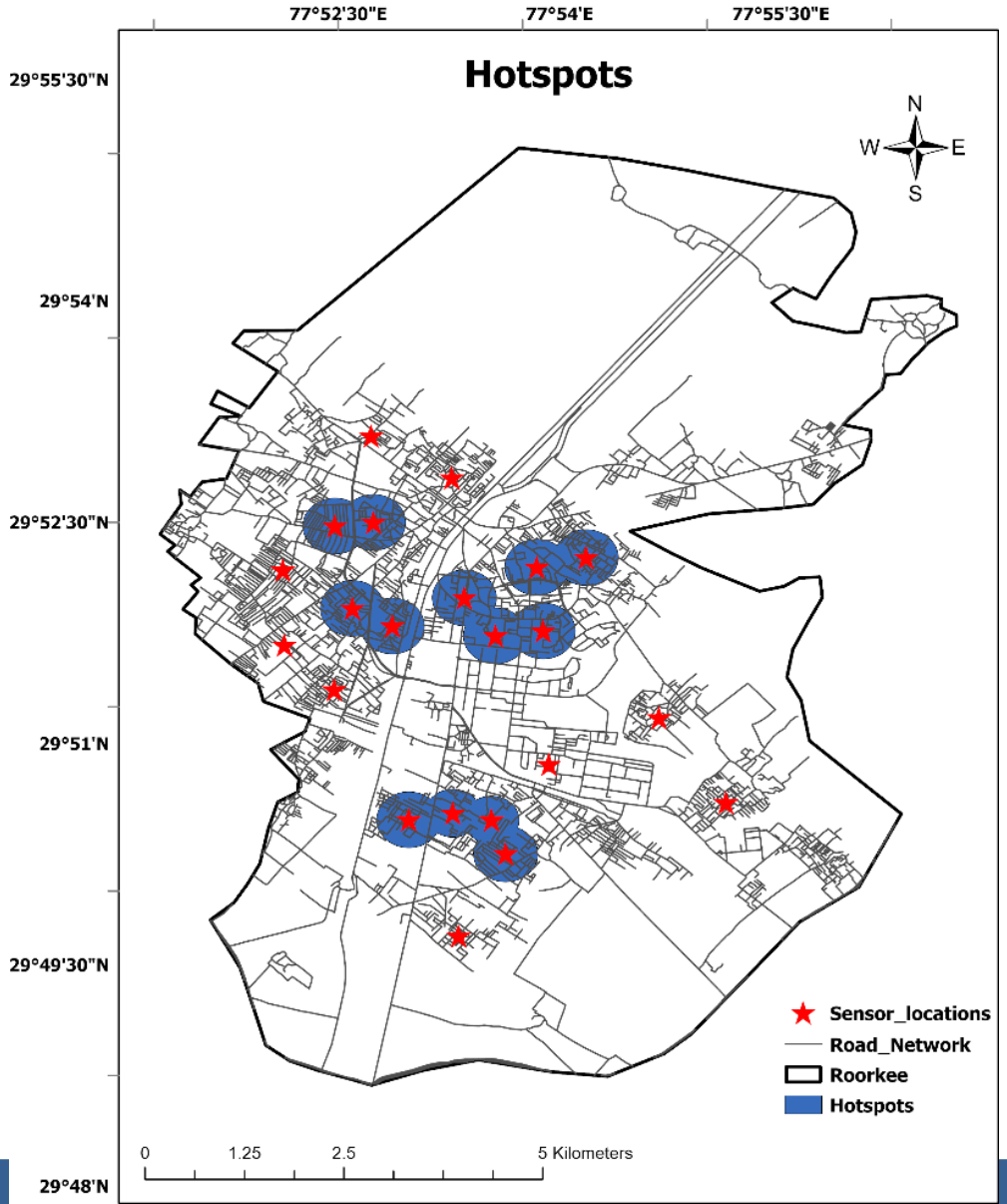


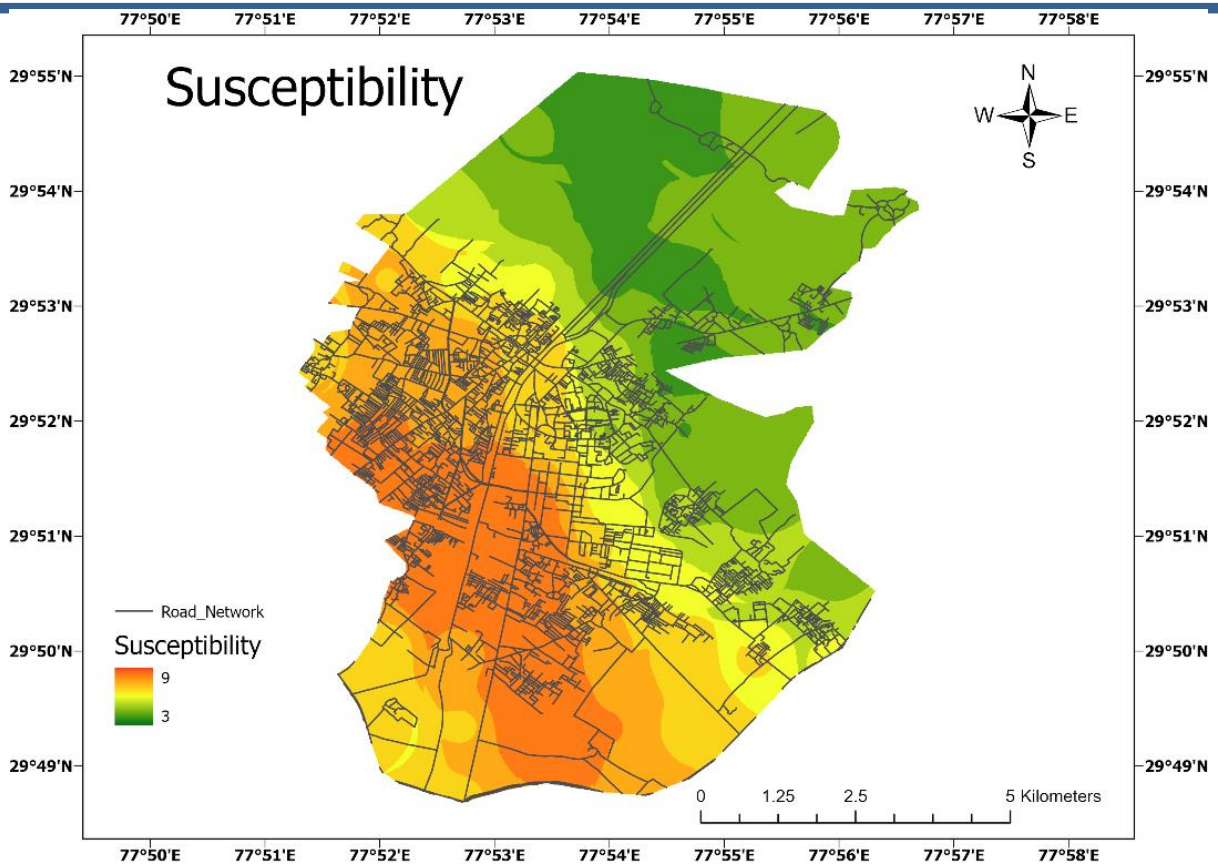
Table shows the sensor readings in the corresponding manner along with the patterned-graph portraying the variation of dust in the micro-climate of the study area. The temporal-spatial data acquisition always sketches a robust pipeline for an accurate analysis for a smooth and proficient policy intervention.

Hotspot Analysis



- Hotspots achieved through combining dust concentration data from IoT sensors with geospatial data from Sentinel-5P.
- Urban areas, industrial zones, and locations prone to frequent dust storms were highlighted as primary hotspots. The analysis revealed that dust concentration peaks during specific periods, particularly in areas with intense construction activity or near highways.

Susceptibility Mapping



- High dust concentration areas are spatially associated particularly along the areas where industrial activity is evident followed by some construction sites and also, along the major traffic corridors.

Findings of Time-Series Analysis

- A gradual upward trend in dust concentrations during weekdays.
- Seasonal peaks observed daily between 8AM and 11AM and again between 5PM and 8PM.
- Weekend effects were noticeable with dust concentrations approximately 15% lower than the value compared to weekdays.

Impact Evaluation and Implications

- The impact evaluation of this study will produce several key outputs and results that reflect both environmental and socio-economic improvements. The primary outputs include:
  - **Reduction in Dust Concentration Levels:** By comparing pre- and post-intervention dust concentration data from both IoT sensors and Sentinel-5P satellite observations, the evaluation will quantify the reduction in dust concentrations across the identified geographic hotspots.
  - **Improvement in Air Quality Index (AQI):** As dust concentration levels drop, the Air Quality Index (AQI) is expected to improve in the targeted regions. The AQI improvements can be mapped geographically to showcase which areas have experienced the most significant air quality gains.

…. Impact Evaluation and Implications

- Health Impact Reduction: The evaluation will assess the correlation between the decline in dust levels and a reduction in respiratory and cardiovascular health incidents in the affected populations. This will be represented through statistical analyses and health data trends, showing fewer hospital admissions or reported cases in hotspots.
- Economic and Productivity Gains: As air quality improves, secondary benefits such as increased worker productivity, reduced healthcare costs, and enhanced living conditions will be evaluated. Surveys and economic indicators will capture these socio-economic impacts, providing a broader view of how dust pollution mitigation translates into tangible community benefits.

• Overall, the findings demonstrate the effectiveness of combining IoT and geospatial approaches for mitigating dust pollution, supporting data-driven policy and intervention strategies.

Conclusion and Way Ahead

- **Integrated Monitoring Works:** The combination of IoT-enabled dust sensors and Sentinel-5P satellite data delivers a robust, real-time, and scalable environmental monitoring system.
- **Actionable Geospatial Insights:** Hotspot and susceptibility maps offer data-driven support for targeted air quality interventions, urban planning, and pollution control policies.
- **Future Expansion Possibilities:** Broader pollutant integration (e.g., PM2.5, ozone, VOCs)AI-driven predictive modeling for early warning systems, replicability across other domains like water pollution, heat mapping, or deforestation
- **Policy & Public Health Impact:** This framework can guide sustainable development, reduce health risks, and support smart city planning initiatives.


THANK YOU!